FINAL REPORT

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1. Missing Value Handling

In this project, missing values were handled using multiple techniques depending on the data type and distribution.

- Numerical features such as *age, annual_income, and credit_score* were imputed using **mean and median values**.
- Categorical features like *gender and employment_type* were handled using **most frequent (mode) imputation**.
- Advanced techniques like **KNN Imputer and MICE** were also applied to improve accuracy.
- A **missing indicator feature** was created to capture missingness patterns.

These methods ensured that no important information was lost and the dataset remained consistent.

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2. Outlier Handling

Outliers were detected and treated using multiple statistical techniques:

- **Z-score method** to detect extreme values
- **IQR (Interquartile Range)** for robust detection
- **Percentile method** to cap extreme values
- **Winsorization** to limit the effect of outliers

Outliers in features like *annual_income, loan_amount, and credit_score* were handled effectively to improve model performance.

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3. Encoding Techniques

Different encoding methods were applied based on feature type:

- **Ordinal Encoding** → education_level
- **Label Encoding** → gender
- **One-Hot Encoding** → region, loan_purpose

This ensured that categorical data was converted into numerical format suitable for machine learning models.

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4. Feature Scaling

To normalize the data and improve model performance, multiple scaling techniques were applied:

- **Standardization (Z-score scaling)**
- **Min-Max Scaling**
- **Robust Scaling**
- **Normalization**

Scaling helped in handling features with different ranges and improved convergence of ML algorithms.

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5. Feature Engineering

New features were created to enhance model understanding:

- **Debt-to-Income Ratio**
- **Average Monthly Transactions**
- **Spending-to-Income Ratio**

These features provided deeper insights into customer financial behavior.

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6. Transformations Applied

To handle skewed data distributions:

- **Log Transformation**
- **Square Root Transformation**
- **Box-Cox and Yeo-Johnson Transformations**

These transformations improved data normality and model performance.

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7. Final Dataset Readiness

After applying all preprocessing steps:

- The dataset is **clean and free of missing values**
- Outliers are properly handled
- All categorical variables are encoded
- Features are scaled and transformed
- New meaningful features are added

The dataset is now **fully prepared and ready for machine learning modeling**.